

class 1

intro

computing background thru PhD and then job providing research computing support in stats/econ/campus

please call me Chris

can I take the course: - fill out survey so I can add you to Ed (all announcements) - no third section, so only 74 - priority - 69 enrolled - 5 spots - Berk grad yes, Berk UG prob, hopefully conc enroll - you should attend and do the work so don't fall behind

243A/B experiment - have had to move some content around so some adjustments relative to previous years jump right into the bash shell on Friday, more quickly than in previous years

logistics

material on website

get software set up this week - help in optional lab on Fri

exam - don't schedule travel

problem sets have been critical to learning; I think working through problems is still essential, but instructors are making changes in light of the power of AI

PS grading is for completion but with in-lab quizzes directly related to the material. My goal is that if you really did the work on the PS, the quiz should be straightforward.

Grading mostly for whether you completed the work not for correctness, but we will mark off for sloppy, poorly-formatted, disorganized, badly-structured code.

AI

AI: spend a bit of time talking about AI assistance, both in terms of how it affects *what* we should be learning in order to do computation effectively in our careers and *how* we should / should not use AI to learn

I don't have all (or even many) of the answers; you all are the ones who have been learning in the past 5 years in the context of AI.

machine - assembly - C - Python - AI coding agent (draw on board)

1. brainstorm why still need Python
2. tinyurl.com/st243-ai: Google form on AI (we'll have regular Google forms during class sessions for class participation - this is not one of them)
3. tinyurl.com/st243-ai2: google form on AI queries

shell

start with the shell as working on the command line is at the core of modern scientific computing, the internet, tech workflows common baseline allows us to do work reproducibly, efficiently and allow for automation I'll say more context on Fri/next week in Unit 2

see unit 1 notes

a little shell exploration

terminal locally terminal + ssh to gandalf vs code terminal

- open terminal
- ssh to gandalf
- clone repo
- git pull
- git log
- git branch

context: - conda env list - source activate unit5 - pwd - whoami

- ls
- ls -l
- ls -a
- ls -all
- ls -al

relative/absolute paths

- pwd

- `cd ../units`
- `cd /accounts/vis/paciorek/Desktop/fall-2026/units`

`man?` - `/` = search - `n` = next - `q` = quit - space = page down

schedule

see schedule

- Friday lab - software help and help if not comfortable with basic work in shell (`cd`, `ls`, `cp`)
- next 2-3 days work on basic shell work if not comfortable with what I just demo'ed
- if have trouble accessing shell don't fall behind - just use `stat.datahub.berkeley.edu` (or `datahub.b.e`)

Monday - bash shell assignment - informal